

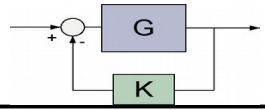
ME 2801

Introduction to Control Systems

W1: Inverse Laplace
Partial Fraction Expansion with Complex Roots



Inv. Laplace w/Complex Roots



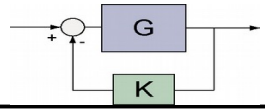
Given $F(s)$

$$F(s) = \frac{1}{(s+1)(s^2+9)}$$

- Find $f(t)$



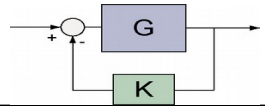
Inv. Laplace w/Complex Roots



Given $F(s)$

$$F(s) = \frac{1}{(s+1)(s^2+9)} = \frac{A}{(s+1)} + \frac{B(s)}{(s^2+3^2)} + \frac{C(3)}{(s^2+3^2)}$$

- Find $f(t)$



Given $F(s)$

$$F(s) = \frac{1}{(s+1)(s^2+9)}$$

- Find $f(t)$

PFE with MATLAB

```
>> num = [1];
den = conv([1 1],[1 0 9]);
[K,p,k]=residue(num,den)
K =
    -0.0500    - 0.0167i
    -0.0500    + 0.0167i
     0.1000    + 0.0000i
p =
     0.0000    + 3.0000i
     0.0000    - 3.0000i
    -1.0000    + 0.0000i
k =
     []
```

Given $F(s)$

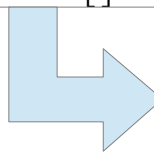
$$F(s) = \frac{1}{(s+1)(s^2+9)}$$

- Find $f(t)$

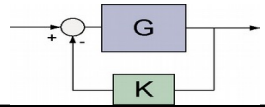
PFE with MATLAB

```
[K,p,k]=residue(num,den)
K =
    α-βi
    α+βi
    ...
p =
    +ωi
    -ωi
    ...
k =
    []
```

```
>> num = [1];
den = conv([1 1],[1 0 9]);
[K,p,k]=residue(num,den)
K =
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    0.0000   - 3.0000i
   -1.0000   + 0.0000i
k =
    []
```



$$F(s) = \frac{2(\alpha)s + 2(\beta)(\omega)}{s^2 + \omega^2} + \dots$$



Given $F(s)$

$$F(s) = \frac{1}{(s+1)(s^2+9)}$$

- Find $f(t)$

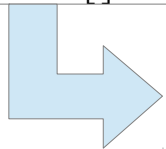
PFE with MATLAB

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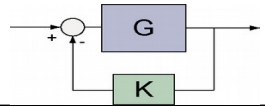
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     0.0000 + 3.0000i
     0.0000 - 3.0000i
    -1.0000 + 0.0000i
k =
     []
```

$$F(s) = \frac{2(-0.05)s + 2(0.0167)(3)}{s^2 + 3^2} + \frac{0.1}{(s+1)}$$

$$F(s) = -0.1 \frac{s}{s^2 + 3^2} + 0.033 \frac{3}{s^2 + 3^2} + 0.1 \frac{1}{(s+1)}$$



$$F(s) = \frac{2(\alpha)s + 2(\beta)(\omega)}{s^2 + \omega^2} + \dots$$



Given $F(s)$

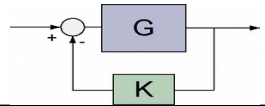
$$F(s) = \frac{1}{(s+1)(s^2+9)}$$

$$F(s) = -0.1 \frac{s}{s^2+3^2} + 0.033 \frac{3}{s^2+3^2} + 0.1 \frac{1}{(s+1)}$$

- Find $f(t)$

$$f(t) = (-0.1) \cos(3t) + (0.033) \sin(3t) + (0.1) e^{-t}$$

- Check by taking Laplace transform of $f(t)$



Given $F(s)$

$$F(s) = \frac{1}{(s+1)(s^2+9)}$$

- Find $f(t)$

PFE with MATLAB

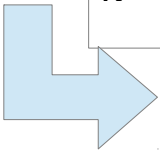
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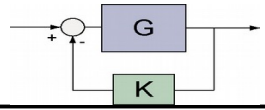
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$$f(t) = (-0.1) \cos(3t) + (0.033) \sin(3t) + (0.1) e^{-t}$$



$$F(s) = \frac{2(\alpha)s + 2(\beta)(\omega)}{s^2 + \omega^2} + \dots$$



For inverse Laplace transform problems with complex (or imaginary) poles (roots of the denominator)

- Use MATLAB `residue()` function for algebra of PFE
- Use handout to interpret `residue()` output
 - Match values for a , ω , α , β

```
[K, p, k] = residue(num, den)
K =
     $\alpha - \beta i$ 
     $\alpha + \beta i$ 
    ...
p =
     $-a + \omega i$ 
     $-a - \omega i$ 
    ...
k =
    []
```